
Contrastive Decoding Permutes, It Does Not Ground

Section draft — CHAIR analysis on LLaVA-1.5-7B and Qwen2.5-VL-7B

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1 Hypothesis

Contrastive decoding rests on a specific mechanistic story: a degraded amateur branch amplifies prior-driven hallucinations, and subtracting its logits from the expert steers generation toward better-grounded tokens. This story implies two properties that should be directly observable at decoding time. We state them as hypotheses and design Experiments 2 and 3 to measure each one on two model families (LLaVA-1.5-7B and Qwen2.5-VL-7B).

H1 (Intervention). Contrastive decoding should change the emitted token at the steps where hallucinations arise. Experiment 2 measures how often the contrastive output coincides with the expert’s own greedy choice, and compares this rate at real- versus hallucinated-object steps, to quantify where and how often the intervention takes effect.

H2 (New grounding). When contrastive decoding does intervene, it should introduce a better-grounded alternative — changing which candidates are available, rather than re-ordering a shortlist the expert has already fixed. Experiment 3 measures the overlap between the expert, amateur, and contrastive candidate sets, to determine whether the contrast supplies new candidates or re-weights existing ones.

Together these experiments characterize the contrastive step mechanistically: Experiment 2 establishes *when* it acts, and Experiment 3 establishes *what* it has available to act on. This complements the external, benchmark-level analysis of Yin et al. [2025] by examining the decoding process itself, on the generative captioning setting measured by CHAIR.

2 Experimental Setup

2.1 What the experiments do

Both are *observational* instrumentations of a normal CD run: the decoding algorithm is unaltered, we only record its internal state at every step.

Exp. 2 (Intervention rate; H1). At each step t we record the token CD emits, y_t , and the expert’s own greedy choice, $\arg \max_w E_t(w)$, and report how often they coincide — over all steps, and restricted to steps emitting a **real** or **hallucinated** object. Agreement means CD changed nothing.

Scope. Since this instruments a real CD decode, E_t is teacher-forced on CD’s own prefix $y_{<t}$. The question is: *holding history fixed at what CD actually generated, would plain greedy differ at this step?* It is not a sequence-level claim about matching an independent greedy rollout. The step-level framing isolates the contrastive step’s contribution at the moment it acts; a sequence-level comparison would conflate it with compounding autoregressive divergence. It is also conservative: any argmax difference counts as an intervention, however inconsequential.

Exp. 3 (Candidate overlap; H2). At each step we record the top-10 under three rankings — expert E_t , amateur A_t , contrastive $C_t = (1+\alpha)E_t - \alpha A_t$ (pre-APC) — and measure (i) **set overlap** (same words, order-insensitive) and (ii) **exact-position match**. The pair separates “do the branches consider the same words?” from “do they order them the same way?”

2.2 Models, data, decoding

Models (both float16, sdpa): **LLaVA-1.5-7B** (CLIP ViT-L/14-336px $\rightarrow$ linear projector $\rightarrow$ Vicuna-7B-v1.5; fixed 576 image tokens; vicuna_v1 template; LLaMA/SentencePiece tokenizer, vocab 32k, word-initial marker `_`) and **Qwen2.5-VL-7B-Instruct** (dynamic-resolution ViT, 2×2 merging, 3D mRoPE; visual-token count is image-dependent, ≈ 266 at `min_pixels=256 · 28 · 28` / `max_pixels=1280 · 28 · 28`; `IMAGE_TOKEN_ID=151655`; chat template; byte-level BPE, vocab 151,643, word-initial sub-tokens carry a leading space).

Data. 500 MSCOCO val2017 images — the *same ids, same order*, for every method and both models (verified), so all comparisons are paired. 500 is the conventional CHAIR sample size [Huang et al., 2024]. We deviate from the prevailing val2014 convention; this shifts absolute CHAIR values but no within-study comparison, since every method sees identical images. Ground truth is standard CHAIR [Rohrbach et al., 2018]: COCO instance-segmentation categories $\cup$ objects named in the five reference captions, mapped to MSCOCO-80 via the CHAIR synonym dictionary. Prompt: “Describe this image in detail.”

Decoding. Greedy (`do_sample=False`), `max_new_tokens = 256`; determinism isolates the contrastive mechanism from sampling noise. Following the reference implementation,

$$C_t(w) = (1 + \alpha)E_t(w) - \alpha A_t(w), \quad \mathcal{V}_{\text{head}} = \{w : E_t(w) \geq \log \beta + \max_{w'} E_t(w')\}, \quad (1)$$

with $\alpha = 1.0$, $\beta = 0.2$; tokens outside $\mathcal{V}_{\text{head}}$ are set to $-\infty$ before the arg max. **The APC cutoff depends on the expert logits only.** The greedy baseline is plain $\arg \max_w E_t(w)$.

VCD. Amateur = diffusion-noised image: $T = 1000$, $\beta_{1:T} = \sigma(\text{linspace}(-6, 6, 1000)) \cdot (0.5 \times 10^{-2} - 10^{-5}) + 10^{-5}$, $\bar{\alpha}_t = \prod_{s \leq t} (1 - \beta_s)$, $v' = \sqrt{\bar{\alpha}_T}v + \sqrt{1 - \bar{\alpha}_T}\epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, at `noise_step=500`; drawn once per image, fixed across that caption’s steps. All §3 numbers use 500; analyses elsewhere in this paper at other noise levels are separate experiments, labelled at point of use.

SID. Reference recipe: `AGG_LAYER=2`, `ATTENTION_RANK=72` of 576 (a *random* 12.5% subset of image tokens is *kept*, drawn once per image); layers 0–1 see the full image, layer ≥ 2 masks all non-kept image tokens. On LLaVA this is the vendored `use_sid` path. On Qwen2.5-VL we reproduce it by per-layer mask injection under sdpa, because eager attention is numerically broken for Qwen2.5-VL in transformers 5.x (it emits degenerate text), making the original eager-mask hook unusable: at layers $\geq \text{AGG_LAYER}$ we inject a causal 4D additive mask setting non-kept image key-columns to $-\infty$. Verified to change amateur logits vs. no masking ($|\bar{\Delta}| = 0.77$ over the expert’s top-30) and to differ from all-layer masking ($|\bar{\Delta}| = 2.01$), confirming layers 0–1 stay unmasked. Qwen applies the same 72/576 fraction of that image’s (dynamic) visual tokens.

2.3 Instrumentation and labelling

At every step we log the top-30 ids and logits for expert, amateur, contrastive (pre-APC) and post-APC scores, plus the APC cutoff, emitted token id and per-candidate APC-survival flag: 60,094/59,271 steps for LLaVA (VCD/SID), 93,652/94,081 for Qwen. We store **ids and logits only** — no token-level semantic labels — so object attribution happens downstream at the caption level, which structurally prevents sub-word mislabelling.

Object attribution. Labels come from standard CHAIR applied to the *generated caption text*, never to individual sub-word tokens: words are singularized, multi-word categories collapsed (“*teddy bear*”, “*dining table*”), and matched to MSCOCO-80 through the synonym dictionary (*man/woman/guy* $\rightarrow$ *person*). A mention is **real** iff its category is in the image’s ground truth, else **hallucinated**. Each mention maps back to the step producing its first token via incremental detokenization.

This discipline is not cosmetic, and we flag it because it changed our own conclusions. Our first pass labelled each decoding step by string-matching the token emitted at it. That matcher scores the “*ship*” inside “*companionship*” as a boat, splits “*bear*” from “*teddy bear*” and counts it as the animal, and silently misses any object whose name the tokenizer splits. Under those labels, LLaVA’s CD-vs-greedy agreement at hallucinated-object steps reads 87.0% (VCD) and 81.3% (SID); under caption-level CHAIR it is 75.1% and 73.1%. The two are not the same quantity measured twice: the per-token pass ranges over 384/364 individual *tokens* tagged by string match, the corrected pass over 595/577 caption-level object *mentions*. The earlier figures were computed on an incoherent unit and

we report them only to document the correction; every number in this paper uses the caption-level procedure, for both models. Qwen2.5-VL was labelled at the caption level from the outset and is unaffected.

2.4 Uncertainty

All intervals are **image-level (cluster) bootstrap** 95% CIs, $B = 10,000$. Steps are not i.i.d.: they nest within captions within images, and images differ systematically (a cluttered image yields both more hallucinations and different agreement behaviour). A per-step Wilson/ z interval ignores this clustering and can understate uncertainty. We therefore resample the 500 **images** with replacement — the item-level unit used elsewhere in this paper — and recompute each statistic from resampled totals; since every quantity is a ratio of sums ($\sum_i \text{hall_eq}_i / \sum_i \text{hall_n}_i$), this correctly weights images by their number of relevant steps. The real-vs-hallucinated gap is tested by the same bootstrap (reported p is the two-sided proportion of resamples where the gap changes sign). Clustering proves to matter little — LLaVA-VCD’s hallucinated-step interval is $[71.4, 78.8]$ against a naive Wilson $[71.5, 78.4]$ — but we report the bootstrap regardless, since its validity does not depend on the effect being large enough to survive the wrong analysis.

2.5 Environment and limitations

Environment. PyTorch 2.8.0+cu128, transformers 5.12.1, qwen-vl-utils; one NVIDIA L4 (24 GB). LLaVA runs both branches in a manual KV-cached loop; Qwen caches the expert via `generate()` but recomputes the amateur *cache-less* each step, because a hand-rolled mRoPE second-branch cache diverged from the exact computation by up to 5.6 logits (mean 2.8) — far beyond fp16 noise — so we kept the correct-by-construction path. Caching is a speed optimization only and does not change computed logits. Checks confirm the amateur is bit-exactly deterministic under a fixed noise draw ($\max |\Delta| = 0$) and that interleaving it does not corrupt the expert (an independent teacher-forced pass reproduces captured expert logits to $|\overline{\Delta}| \approx 0.5$ on logits of magnitude 15–25, the known fp16 cached-vs-uncached tolerance; cached values are the ecologically valid ones, since deployed CD runs cached). **No global seed was fixed:** VCD’s noise draw and SID’s kept subset are stochastic by construction, as in the reference implementations, so per-image realizations differ across runs. All quantities aggregate over $\geq 59,000$ steps on the same fixed 500 images, a regime where this variance is negligible against the reported effects.

Limitations. (1) Logit statistics use the top-30, not the full vocabulary — exact for APC-survivor counts (APC retains 1–2 tokens, always within the top-30) but candidate-level statistics ignore the tail. (2) CHAIR’s ground truth is incomplete: an object present but neither segmented nor mentioned in any reference caption scores as a hallucination. (3) Only MSCOCO-80 categories are scored, since ground truth exists only for them. All three apply *identically* to every method and both models, so they cannot bias the paired comparisons carrying our claims.

3 Results

All intervals are image-level bootstrap 95% CIs (§2.4).

3.1 Experiment 2: CD reproduces the undefended choice at most hallucination steps

Table 1 reports how often CD’s emitted token is the expert’s own greedy choice, holding the prefix fixed.

(i) *The contrastive step is a no-op at most steps.* On LLaVA 94.0% (VCD) and 91.8% (SID) of emitted tokens are identical to the undefended model’s: the mechanism is inactive for roughly nineteen of every twenty tokens.

(ii) *Most hallucinations are emitted unchanged.* At steps emitting a hallucinated object CD still reproduces the expert’s greedy choice 75.1% / 73.1% of the time on LLaVA and 62.2% / 62.5% on Qwen. A hallucination emitted with the very token the undefended model would have emitted was not prevented by CD. This is a hard ceiling: CD can be causally responsible for at most $\approx 25\%$ (LLaVA) or $\approx 38\%$ (Qwen) of hallucination-step outcomes, and only if every intervention corrects rather than introduces an error. Even at the pessimistic bootstrap ends (78.8%, 68.0%) the majority

Table 1: **Intervention rate.** % of steps where CD’s output equals the expert’s own greedy top-1 (i.e. CD changed *nothing*). Object columns restrict to steps emitting a CHAIR-labelled mention; “Gap” is real – hallucinated (pp). Image-level bootstrap 95% CIs ($B = 10,000$, 500 images); every gap excludes zero ($p < 10^{-4}$).

Model	Method	All steps	Real-object	Hallucinated-object	Gap (pp)
LLaVA-1.5-7B	VCD	94.0 [93.8, 94.2]	92.7 [91.8, 93.6]	75.1 [71.4, 78.8]	17.6 [13.8, 21.4]
	SID	91.8 [91.6, 92.1]	90.7 [89.7, 91.7]	73.1 [69.5, 76.7]	17.6 [14.0, 21.3]
Qwen2.5-VL-7B	VCD	75.3 [74.9, 75.7]	83.7 [82.1, 85.3]	62.2 [56.3, 68.0]	21.5 [15.6, 27.6]
	SID	75.7 [75.3, 76.1]	84.6 [83.1, 86.1]	62.5 [56.5, 68.1]	22.1 [16.3, 28.1]

Table 2: **Top-10 set overlap** (order-insensitive) of the expert top-10 with the amateur ($E \cap A$) and contrastive ($E \cap C$) top-10, out of 10, reported over all steps and restricted to steps emitting a real- or hallucinated-object mention. E : expert, A : amateur, C : $2E - A$ (pre-APC). Image-level bootstrap 95% CIs ($B = 10,000$); step counts (all/real/hall): LLaVA $\sim 60k/3.2k/0.6k$, Qwen $\sim 94k/2.6k/0.3k$.

Model	Method	All steps		Real-object		Hallucinated	
		$E \cap A$	$E \cap C$	$E \cap A$	$E \cap C$	$E \cap A$	$E \cap C$
LLaVA-1.5-7B	VCD	9.34 [9.32, 9.36]	9.33 [9.31, 9.35]	8.89 [8.84, 8.94]	8.92 [8.88, 8.97]	8.42 [8.30, 8.54]	8.34 [8.21, 8.47]
	SID	9.14 [9.12, 9.15]	9.10 [9.09, 9.12]	8.71 [8.67, 8.75]	8.68 [8.64, 8.72]	8.35 [8.25, 8.45]	8.06 [7.94, 8.17]
Qwen2.5-VL-7B	VCD	8.36 [8.34, 8.38]	8.01 [7.98, 8.04]	8.03 [7.97, 8.10]	7.93 [7.87, 7.99]	7.72 [7.50, 7.93]	7.78 [7.60, 7.94]
	SID	8.20 [8.18, 8.23]	7.95 [7.92, 7.98]	7.83 [7.77, 7.89]	7.86 [7.80, 7.92]	7.37 [7.17, 7.58]	7.56 [7.38, 7.72]

are emitted unchanged. H1 fails: a method advertised as suppressing hallucinated tokens mostly agrees with them.

(iii) *Yet CD is selectively active — which strengthens the finding.* The obvious objection is that CD “just never fires,” rendering this trivial. It does not hold: agreement is 17.6–22.1 pp *lower* at hallucinated- than real-object steps in all four settings, every CI excluding zero. The signal is concentrating where intended. The mechanism fires and does not deliver, which forecloses the first repair a reader reaches for (“tune α so it fires more”). Firing more is not the missing ingredient (§3.2).

(iv) *The models fail differently.* Qwen’s overall agreement (75%) is far below LLaVA’s (94%) — CD is $\approx 4\times$ as interventionist — yet its CHAIR scores degrade rather than improve. An architecture where CD rewrites one token in four and still worsens hallucination refutes the “more contrast = less hallucination” intuition, and shows the effect is not a LLaVA-specific artifact.

3.2 Experiment 3: all three branches deliberate over the same words

Table 2 inspects the candidate sets themselves.

The amateur proposes the same words. Across all steps, 9.34 (VCD) / 9.14 (SID) of the expert’s top-10 also appear in the amateur’s on LLaVA, and 8.2–8.4 on Qwen. The overlap stays high even where it should not: at hallucinated-object steps it is still 8.3–8.4 of 10 on LLaVA and 7.4–7.7 on Qwen. The amateur — viewing an image destroyed by diffusion noise, or 12.5% of its tokens — proposes almost the same candidates as the expert, including at the steps that emit a hallucination. The difference $d = E - A$ therefore cannot isolate an ungrounded candidate, because scarcely any candidate is considered by only one branch; what d carries is a re-weighting of a shortlist the expert has already fixed.

The contrast supplies no new candidates. The contrastive top-10 overlaps the expert’s by essentially the same amount as the amateur does: $E \cap C$ tracks $E \cap A$ in every cell of Table 2 (e.g. 9.33 vs. 9.34 on LLaVA-VCD, 8.01 vs. 8.36 on Qwen-VCD). A better-grounded word the expert never short-listed is thus not brought in by the contrast; it stays outside the candidate set. The adaptive plausibility constraint reinforces this, gating on the *expert* logit alone: any token below $\log \beta + \max_w E_t(w)$ is set to $-\infty$ regardless of its contrastive score. The expert’s own ranking selects the pool the method may choose from. This is H2 failing: no new grounding.

Table 3: **Object-shortlist composition at hallucination steps.** At steps that emit a hallucinated object, the top-10 candidates are classified into real-object, hallucinated-object, and non-object tokens; among the object candidates only, we report the % that are real (R) and the % that are hallucinated (H), which sum to 100, averaged over all such steps.

Model	Method	Expert		Amateur		Contrastive	
		%R	%H	%R	%H	%R	%H
LLaVA-1.5-7B	VCD	16.8	83.2	16.9	83.1	17.6	82.4
	SID	17.7	82.3	17.7	82.3	17.5	82.5
Qwen2.5-VL-7B	VCD	16.0	84.0	18.7	81.3	13.7	86.3
	SID	14.5	85.5	20.1	79.9	14.2	85.8

Why the models differ. The one place the candidate set does move is Qwen. LLaVA’s contrast leaves the expert’s set almost untouched ($E \cap C = 9.33$ of 10, barely below $E \cap A$), so at the set level it is close to a null operation. Qwen’s contrast removes about two of the expert’s ten candidates ($E \cap C = 8.01$, itself below $E \cap A = 8.36$): it disrupts the candidate set more, yet the substitutions are no better grounded, since overlap with the amateur stays high. This resolves Exp. 2’s puzzle — why Qwen’s CD intervenes $\approx 4\times$ as often while making CHAIR worse: changing the candidate set more, without adding grounding, is disruption without benefit — small enough on LLaVA to be invisible, large enough on Qwen to hurt.

3.3 Composition of the object shortlist at hallucination steps

The overlap results show the branches share a candidate set; this measurement asks what that set is *made of* at the moment a hallucination is produced.

Method. We restrict to steps that emit a hallucinated object (labelled at the caption level, §2.3). At each such step we take a branch’s top-10 candidate tokens and classify each as a real-object, a hallucinated-object, or a non-object token. The emitted token uses its known CHAIR label; every other candidate is decoded to a word, kept only if it is word-initial, singularised, and matched to an MSCOCO-80 category through the CHAIR synonym dictionary — real if that category is in the image’s ground truth, hallucinated otherwise, and non-object if it matches nothing. We then discard the non-object tokens and, *among the object candidates only*, report the share that are real versus hallucinated, averaging over all hallucination-emitting steps (each object candidate weighted equally). Because a single token cannot resolve multi-word categories (e.g. “stop” from “stop sign”), such candidates are conservatively counted as non-objects; this applies equally to every branch and model.

Table 3 reports the result. When the model emits a hallucination, the object candidates it is choosing among are overwhelmingly *other* hallucinations: 82–86% across both models, leaving only about one object candidate in six that is real. This composition is set by the expert before the contrast is applied, and contrastive decoding does not repair it — on Qwen it makes the real share *smaller* (VCD 16.0% $\rightarrow$ 13.7%; SID 14.5% $\rightarrow$ 14.2%). The correct object is scarce in the shortlist precisely when it is needed, and the contrast cannot introduce what the shortlist lacks.

3.4 Synthesis

Exp. 2 shows CD does not intervene on most hallucinations though it aims at them; Exp. 3 shows why better aim would not help — the contrast operates inside a pool the expert fixed and the APC narrows using the expert’s logits alone. Together they support more than “CD’s gains are small”: they give the structural reason the gains *must* be small. A method that can only permute the undefended model’s shortlist cannot add grounding that shortlist lacks. Reported CHAIR improvements from such a mechanism are better explained as re-ranking variance than hallucination suppression — corroborated by our proxy experiments, where magnitude-matched random noise reproduces LLaVA’s CD behaviour.

Two qualifications. Object attribution is bounded by CHAIR’s MSCOCO-80 ground truth (§2.5); hallucination-step counts (294–595 per setting) are the smallest samples here, hence the bootstrap

intervals. And these experiments characterize the decoding-time *mechanism*; end-task quality is addressed by our CHAIR scores separately.

References

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